

222862876 KUNENE TS

221571614 MABASO T.S

222871603 Ngwenya LM

222889707 Zulu KS

230522413 Mohuba B

231735305 LISSOKO OK

224870809 DIPHOFA KT

220207870 MTSHALI PK

218232388 Chikong ANCD

217062780 Mamakoko L

220236498 Makama TF

N.B Greetings Sir, Please note that all those who are highlighted with red did not contribute towards Assessment 1.

eVoting System

Cryptographic Voting Website

Trust by Design: Cryptographic Voting Website with AI Voter Assistance

1. SYSTEM OBJECTIVES

This project aims to develop a secure cryptographic voting website for national-level elections. The system ensures:

- Secure voter authentication via Mock Home Affairs database
- One-person-one-vote enforcement
- Encrypted vote storage
- Verifiable vote receipt system
- AI-assisted voter support

The system will be developed using:

- Java (Spring Boot)
- MySQL
- HTML/CSS/JavaScript
- RSA encryption & SHA-256 hashing

2. USERS / ACTORS

1. Voter
2. Election Administrator
3. System Auditor

3. FUNCTIONAL REQUIREMENTS

The system must:

1. Verify voter identity using Mock Home Affairs database
2. Register eligible voters
3. Authenticate voters securely
4. Prevent duplicate voting
5. Encrypt votes before storing
6. Allow administrator to view votes securely
7. Allow auditor to verify vote integrity
8. Provide AI voter assistance
9. Generate election reports

4. NON-FUNCTIONAL REQUIREMENTS

- System must encrypt votes using RSA (2048-bit)
- System must generate SHA-256 receipt hashes
- System must prevent SQL injection
- System must respond within 3 seconds per request
- System must maintain voter privacy
- System must log all administrative actions
- System must run on Java 17

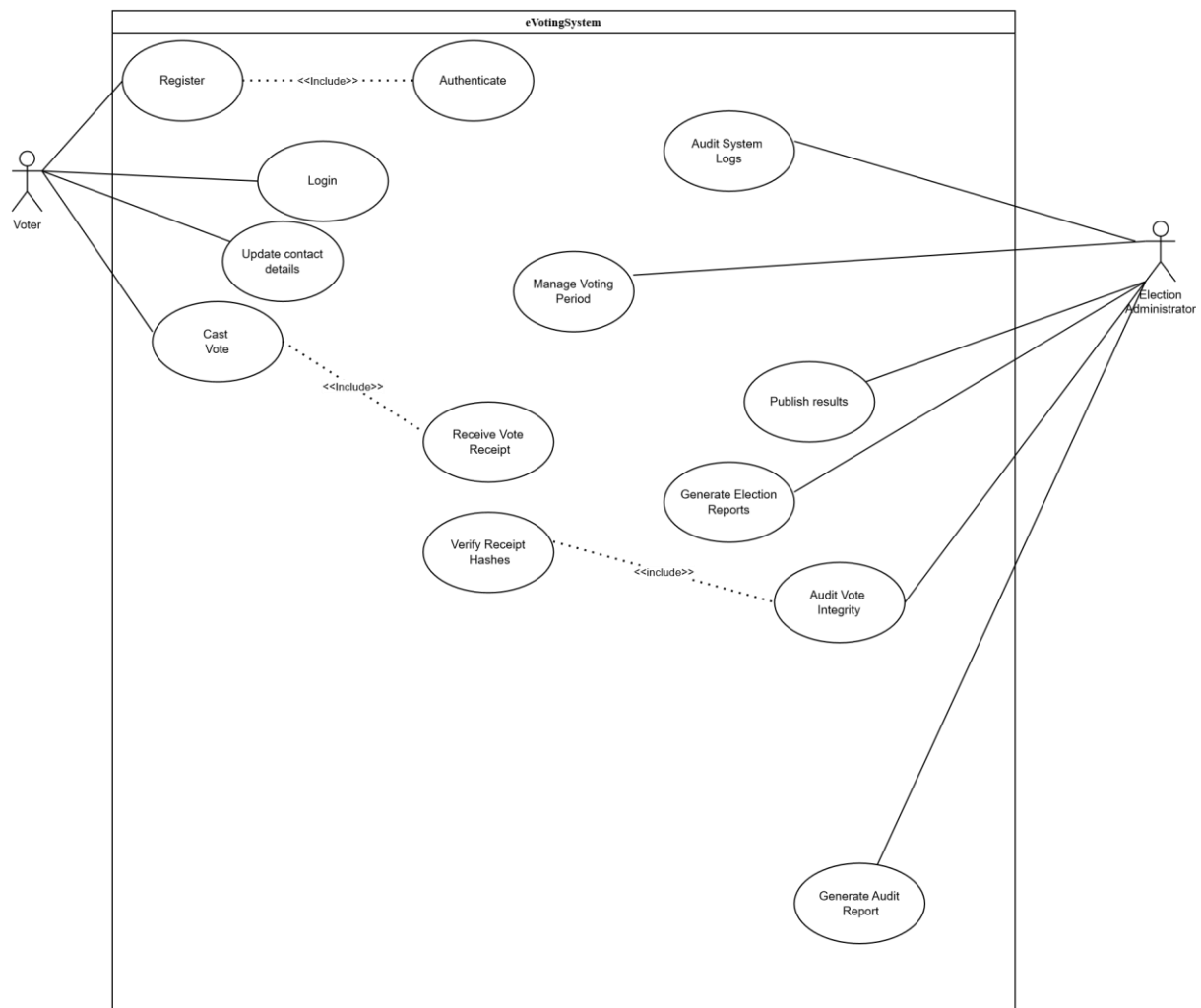
5. EVENT TABLE

Event	Trigger	Source	Use Case	Response	Destination	Action
Voter registers		ID	Voter	identity	Verify voter eligibility	Confirm
Voter approved		ID verified	System	Register voter	Create voter record	Database
Voter logs in		Login attempt	Voter	Authenticate voter	Grant access	Voter
Voter casts vote		Submit vote	Voter	Encrypt vote	Store encrypted vote	Database
Vote recorded		Vote stored	System	Generate receipt	Display receipt hash	Voter
Voter verifies vote		Enter receipt	Voter	Verify receipt	Confirm vote exists	Voter
Admin tallies		Admin request	Administrator	Decrypt votes	Generate results	Admin votes
Admin requests		Admin request	Administrator	Generate election	Print report	Admin report

Auditor checks	Auditor	Audit encrypted	
		Integrity report	Auditor integrity request
votes			
AI assistance	Voter asks	Provide AI	
		Display answer	Voter request question
response			
Double voting	Voter tries	Check voting status	Reject vote Voter attempt
again			

Event	Trigger	Source	Use Case	Response	Destination
Admin views logs	Admin request	Administrator	View audit logs	Display logs	Admin

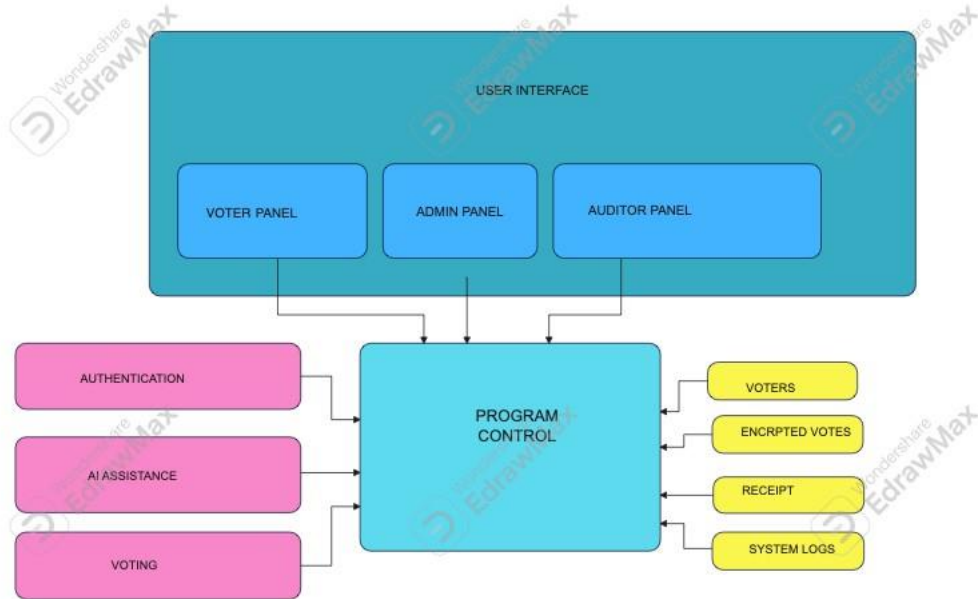
6. USE CASE



Business Process



Block Diagram



N.B Greetings Sir, Please note that all those who are highlighted with red did not contribute towards Assignment 1. Some even said they will do WBS, that is why it is missing in the assessment.